

# Liquid stability, LLE and VLLE

After midterm · Session 3 of 6 · 180 minutes

Soorathep Kheawhom

30 September 2026

# Session outcome

Use global Gibbs-energy support to assess candidate liquid phases.

**Before class:** Review gmix, local curvature and the distinction between stationary and globally stable states.

Learning path and all labs

# The 180-minute class

| IN CLASS                           | MINUTES |
|------------------------------------|---------|
| Recall and prediction              | 10      |
| Concepts and derivation            | 45      |
| Worked example                     | 30      |
| Break                              | 10      |
| Instructor lab demonstration       | 25      |
| Guided student exploration         | 35      |
| Discussion and interpretation      | 15      |
| Exit question and independent task | 10      |

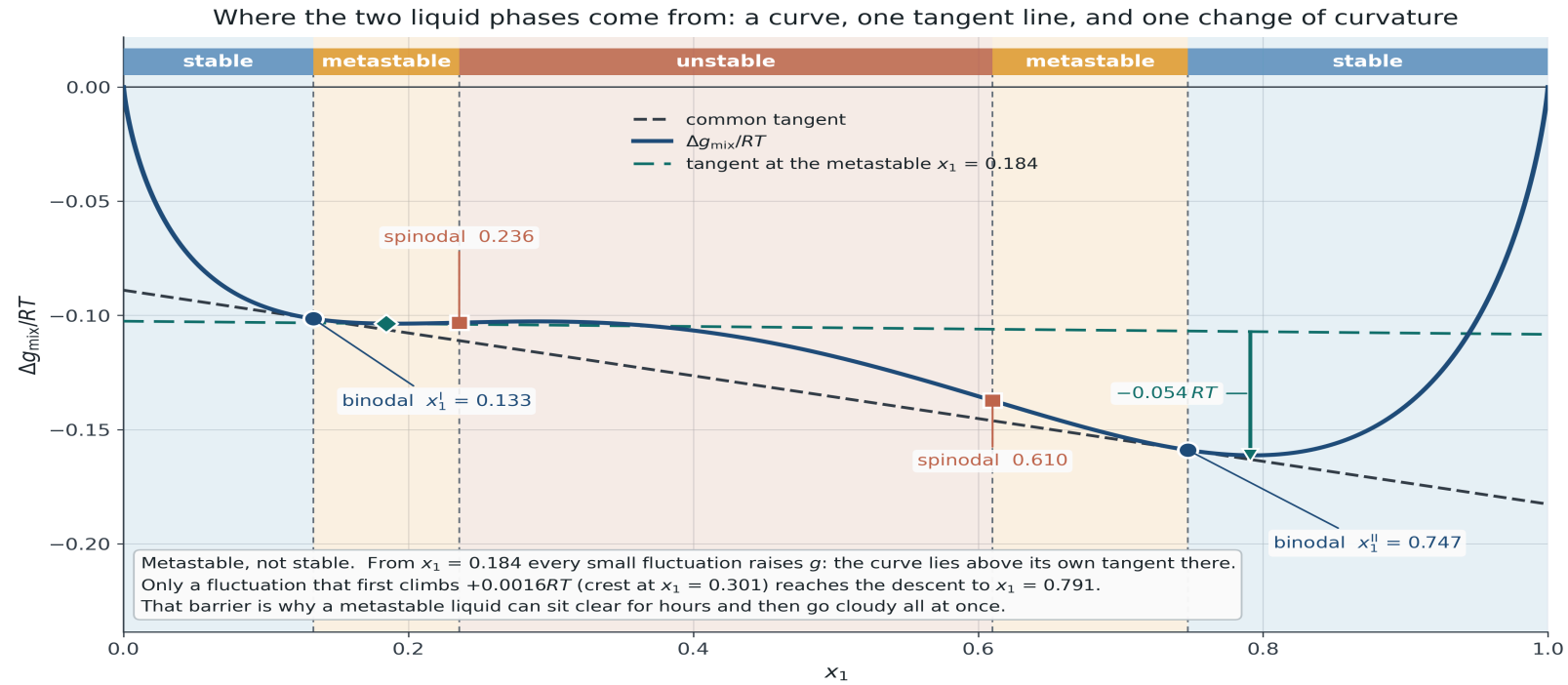
# Phase equilibrium includes stability

Equality of chemical potentials identifies stationary candidate states. Stable equilibrium requires the lowest admissible total Gibbs energy.

$$g_{mix}/RT = \sum_i x_i \ln x_i + g^E/(RT)$$

A split can lower G even when a homogeneous liquid is locally stable.

# Binodal and spinodal



ILLUSTRATIVE PARAMETERS: three-suffix Margules with  $A_{12} = 2.6$ ,  $A_{21} = 2$ , chosen to place a partially miscible binary on one screen. They are not fitted to any measured mixture. Every number printed above is computed from that one  $G^E$ : the binodal by `vlekit.flash.miscibility_gap`, the spinodal by `vlekit.flash.spinodal`, the barrier and the descent by differencing the curve against its own tangent.

Module 5 reference illustration: coexistence and local instability are different boundaries.

# The common-tangent conditions

$$g'(x^\alpha) = g'(x^\beta) = \frac{g(x^\beta) - g(x^\alpha)}{x^\beta - x^\alpha}$$

The tangent must not lie above  $g(x)$  anywhere in the admitted composition interval.

A trivial root  $x^\alpha = x^\beta$  does not establish two distinct liquid phases.

# Tangent-plane distance

For a trial composition  $w$  relative to reference  $z$ ,

$$TPD(w; z) = g(w) - g(z) - g'(z)(w - z)$$

A negative value identifies a Gibbs-energy-lowering perturbation. Local curvature examines only the neighborhood of  $z$ .

Lab 04 evaluates the binary model; it is not a general multicomponent TPD package.

# A lever-rule calculation

Synthetic tie line:  $x_1^\alpha=0.2$ ,  $x_1^\beta=0.8$ . Feed  $z_1=0.5$ .

$$f_\beta = \frac{z_1 - x_1^\alpha}{x_1^\beta - x_1^\alpha} = 0.5$$

For  $z_1=0.35$ ,  $f^\beta=0.25$ . At the same coexistence conditions, the phase amounts change but the tie-line endpoints do not.

# Model shape and liquid splitting

Positive activity deviations alone do not guarantee LLE. The mixing Gibbs-energy shape must support a split.

The labs include Margules and NRTL options with different shapes. Standard Wilson behavior is discussed in the reference deck, not implemented as an LLE solver here.

# **LLE fitting uses different evidence**

Tie-line data constrain the coexistence compositions. Fitting must reproduce distinct phases and a supported common tangent.

A small tie-line error at one  $T$  does not establish a unique parameter set or a temperature law.

Lab 05 retains the input observations and checks stability of the fitted result.

# Class demonstration • Labs 04 and 05

## Stability and LLE • LLE fitting

1. Compare convex and nonconvex mixing Gibbs curves.
2. Inspect global support and the phase fractions.
3. Fit a tie-line example at its stated temperature and inspect the stability check.

# Guided exploration • 35 minutes

Choose one state with a supported liquid split.

Predict the effect of changing  $z$  while holding  $T$  and model fixed. Record phase compositions and amounts, then check the lever rule.

In Lab 05, bootstrap synthetic replicate tie lines at fixed  $T$  and  $\alpha$ . Compare parameter spread with endpoint spread.

# Vapor–liquid–liquid coexistence

$$\hat{f}_i^V = \hat{f}_i^{L\alpha} = \hat{f}_i^{L\beta}$$

For a binary nonreacting system with three phases, the phase rule gives one intensive degree of freedom before imposing T or P.

Phase compositions and phase amounts are different unknowns. Specifying T and P does not always uniquely determine all amounts.

# Three phase amounts can form a family

Given three fixed binary phase compositions,

$$f_{\alpha} + f_{\beta} + f_V = 1$$

$$z_1 = f_{\alpha} x_1^{\alpha} + f_{\beta} x_1^{\beta} + f_V y_1$$

There are two independent constraints on three fractions. Nonnegativity bounds the feasible family.  
Another independent extensive constraint may select a member.

# VLE extension

## Open Lab 06

Compare the supported liquid and vapor states. Move within the admissible phase-fraction family and verify the overall balance.

Identify a state where the vapor fraction is constrained by nonnegativity. Explain why equal fugacities alone cannot determine every fraction.

# Where the model can be reused

A fitted liquid model needs a matching component order, parameter convention and temperature before reuse.

VLE also requires pure-component vapor pressures and a vapor model. LLE experimental pressure is not a vapor-pressure correlation.

A new phase calculation brings new assumptions and validation needs.

# Exit question

A solver reports two liquid compositions with equal chemical potentials and a tiny residual.

What checks distinguish a real stable split from a trivial or metastable result?

Which quantities change when only the overall composition changes?

# Independent practice • suggested 60–90 minutes

Assessment set 3 and rubric • [Offline package](#)

A common-tangent result, one lever-rule check and one limitation of the fitted parameters.

Retain the calculator export, your worksheet, a comparison plot/table and one independent check.

State an assumption that limits your conclusion.

Use the core labs on the [learning path](#). Optional extensions are additional work.

# References and further study

[Module reference deck](#) · [Lab sources and equations](#)

Derivations and original figure references remain in the corresponding module deck. Each lab records its implemented equations and assumptions.

Synthetic worked examples illustrate calculations; they are not evidence of real-system accuracy.